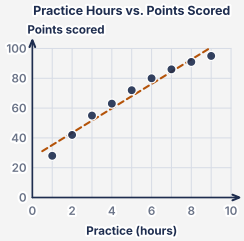
✏️ Unit 7 Practice 1 — Data Analysis

Work through with your teacher · NC.8.SP.1–4

ANSWER KEY

Describe correlation: Up or down? Tight or spread out? Any outliers?

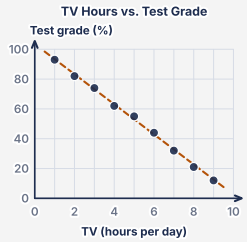
Question 1 Score 3



Describe the correlation in this scatterplot.

- ● **Strong positive correlation**
- ○ Weak negative correlation
- ○ No correlation
- ○ Strong negative correlation

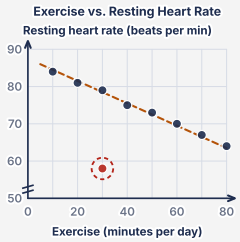
Question 2 Score 3



What type of correlation does this scatterplot show?

- ○ Strong positive
- ○ Weak positive
- ● **Strong negative correlation**
- ○ No correlation

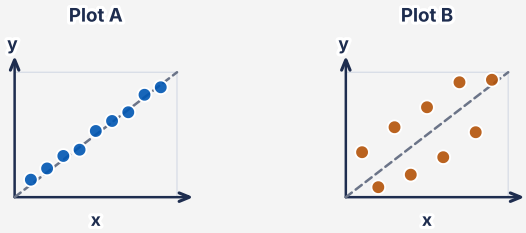
Question 3 Score 4



The **red point** in the scatterplot is an outlier. What is the best description of its position?

- ○ It is the most accurate data point
- ● **Far from the trend — does not follow the overall pattern**
- ○ It is below the x-axis
- ○ It shows the average value

Question 4 Score 4

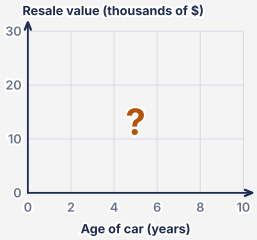


Both trend lines are the same. Look at how close the dots are to the line.

Which statement describes a *strong* correlation vs. a *weak* correlation?

- ○ Strong: the slope is steep · Weak: the slope is shallow
- ● **Strong: points cluster tightly around the trend line · Weak: points are spread far from the trend**
- ○ Strong: there are more data points · Weak: there are fewer points
- ○ Strong: the line goes up · Weak: the line goes down

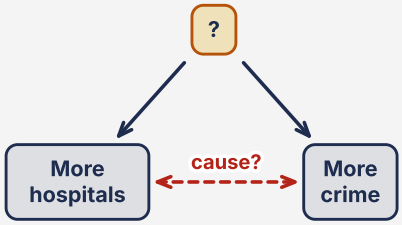
Question 5 Score 4



A scatterplot for "age of car vs. resale value" would most likely show:

- ○ Strong positive correlation
- ● **Strong negative correlation**
- ○ No correlation
- ○ Perfect positive correlation

Question 6 Score 5



Cities with more hospitals also have more crime. Could something else make both go up?

A student notices that cities with more hospitals also have higher crime rates. She concludes: "Hospitals cause crime." What is the flaw in her reasoning?

- ○ Her data must be wrong
- ● **Correlation does not imply causation — larger cities have both more hospitals AND more crime**
- ○ She is correct — the hospitals attract criminals
- ○ She needs a stronger scatterplot

Question 7 Score 5



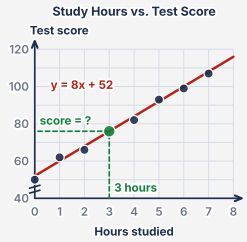
Lower reaction time = faster.

Two students both plot "hours of sleep vs. reaction time." Student A's dots fall in a tight line going down. Student B's dots show very little pattern. **Lower time = faster.** Which student's data best shows that more sleep makes reaction time faster?

- ○ Student B — the weak correlation is more convincing
- ● **Student A — strong negative correlation means more sleep → faster reaction (lower time)**
- ○ Student A — but only if the slope is positive
- ○ Student B — weak correlations are more reliable

y = mx + b: m = slope (rate of change per unit of x) · b = y-intercept (starting value when x = 0)

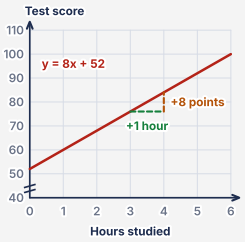
Question 1 Score 3



Using y = 8x + 52: what score is predicted for a student who studies **3 hours**? (Find the green dot.)

- ○ 60 points
- ● **76 points**
- ○ 84 points
- ○ 52 points

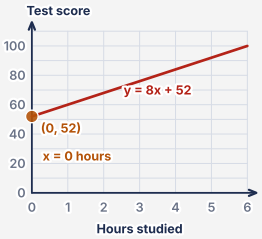
Question 2 Score 3



In the context of *Study Hours vs. Test Score* (y = 8x + 52), what does the **slope of 8** mean?

- ○ A student starts with 8 points
- ● **For each additional hour of study, the predicted score increases by about 8 points**
- ○ The maximum possible score is 8
- ○ The score improves by 52 for each hour

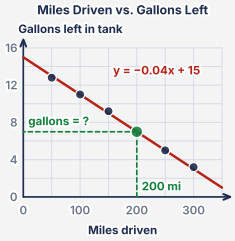
Question 3 Score 4



In the same equation y = 8x + 52, what does the **y-intercept of 52** mean?

- ○ 52 is the highest possible score
- ● **A student who studies 0 hours is predicted to score about 52 points**
- ○ The student gains 52 points per hour
- ○ 52 hours of study is required to pass

Question 4 Score 4



Using y = −0.04x + 15 (x = miles driven, y = gallons remaining): how many gallons remain after 200 miles? (Find the green dot.)

- ○ 11 gallons
- ● **7 gallons**
- ○ 15 gallons
- ○ 3 gallons

Question 5 Score 5



Data was collected for x = 2 to x = 8. A student uses the equation to predict y at x = 25. Is x = 25 inside or outside the data? Is this prediction safe?

- ○ Inside the data — safe
- ● **Outside the data — less safe**
- ○ Inside the data — because x = 25 is a real number
- ○ Neither — predictions are always accurate

Question 6 Score 5



The line of best fit is y = 5x + 10. Data was collected for x = 1 to x = 9. Which x-value is INSIDE the data, so it gives the safest prediction?

- ○ x = 15
- ● **x = 5 (inside the data range 1–9)**
- ○ x = 0
- ○ x = 20

Joint frequency = interior cells · **Marginal frequency** = row/column totals · **Relative frequency** = part ÷ a total. Use the grand total for everyone, or a row or column total for one group.

📋 Use this table for Questions 1–8 (Practice 1)

80 students were surveyed: Do you prefer **Morning** or **Evening** classes? Results split by class period:

	Morning 🌅	Evening 🌙	Row Total
Period 1–3	**28**	12	**40**
Period 4–6	18	22	**40**
Column Total	**46**	**34**	**80**

Question 1 Score 3

	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

Find where the Period 1–3 row and the Morning column cross.

What is the **joint frequency** for Period 1–3 students who prefer Morning?

- ○ 40
- ● **28**
- ○ 46
- ○ 12

Question 2 Score 3

	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

Marginal = the total at the edge of the Evening column.

What is the **marginal frequency** for all students who prefer Evening?

- ○ 22
- ● **34**
- ○ 12
- ○ 40

Question 3 Score 4

	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

Period 1–3: $\frac{28}{40} = ?$

What percent of **Period 1–3** students prefer Morning? (Show calculation)

- ○ 35%
- ● **70%**
- ○ 46%
- ○ 57.5%

Question 4 Score 4

	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

Period 4–6: $\frac{18}{40} = ?$

What percent of **Period 4–6** students prefer Morning?

- ○ 55%
- ○ 22.5%
- ● **45%**
- ○ 62.5%

Question 5 Score 4

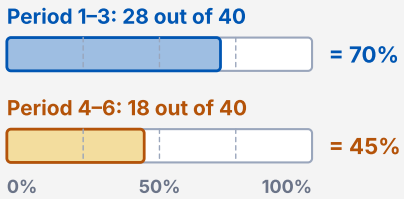
	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

Evening: $\frac{34}{80}$ = ?

What is the **relative frequency** of students who prefer Evening (out of 80 total)?

- ● **42.5%**
- ○ 27.5%
- ○ 55%
- ○ 22.5%

Question 6 Score 5



Based on Questions 3 and 4 (70% vs 45%), is there an **association** between class period and morning preference?

- ○ No — both groups have the same preferences
- ● **Yes — Period 1-3 students strongly prefer Morning compared to Period 4-6**
- ○ No — the row totals are both 40
- ○ Yes — but only because the sample is small

Question 7 Score 5

	Morning	Evening	Total
Period 1–3	28	12	40
Period 4–6	18	22	40
Total	46	34	80

The cell where Period 1–3 meets Evening, out of everyone (the grand total).

If a student is picked at random, what is the probability they are in Period 1-3 AND prefer Evening?

- ● **$^{12}/_{80}$ = 15%**
- ○ $^{34}/_{80}$ = 42.5%
- ○ $^{12}/_{40}$ = 30%
- ○ $^{22}/_{80}$ = 27.5%

📝 Study Tools

📝 Worked Examples

Click "Show Next Step" to reveal solutions one step at a time.

Two-way table: 45 boys like pizza out of 120 total

Step 1: Joint relative freq

```
45 ÷ 120 = 0.375 = 37.5%
```

	Like pizza	Do not like	Total
Boys	45	15	60
Girls			60
Total			120

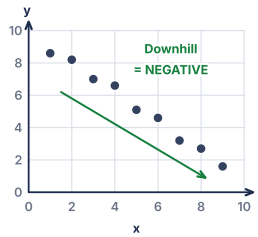
Joint: $\frac{45}{120} = 37.5\%$

Show Next Step →

🔍 Common Mistakes

Can you spot what went wrong?

Correlation: points go up-left to down-right



✗ **Student:** Student says: positive correlation

✓ **Fix:** WRONG! Downhill = NEGATIVE correlation. Uphill = positive.

💡 *Think of it like a hill: going up = positive, going down = negative.*

⚡ Key Tips for This Unit

✅ Positive: both increase. Negative: one up, other down. None: random.

✅ Line of best fit: approx equal points above and below.

✅ Outlier: far from the pattern. Can distort the trend line.

✅ Joint RF = cell÷grand total. Conditional RF = cell÷row or column total.